Task 04 - Image-to-Image Translation using Pix2Pix (cGAN)

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Objective

Implement image-to-image translation using a pre-trained Pix2Pix conditional GAN.

Introduction

Pix2Pix is a conditional GAN that learns mappings between paired images. It contains a Generator and a Discriminator and is widely used for image-to-image translation tasks.

Technologies

Python, Google Colab, PyTorch, Torchvision, Matplotlib, Pillow, Dominate, Visdom.

Workflow

Cloned the official repository, downloaded the pretrained facades model, downloaded the facades dataset, executed inference, and viewed generated outputs.

Implementation

Used the official Pix2Pix implementation and pretrained generator to translate semantic facade labels into realistic building photographs.

Results

Generated images preserved structural information while producing visually realistic building photos. Input, generated, and ground-truth images were compared.

Conclusion

Successfully demonstrated image-to-image translation using Pix2Pix without retraining the network, gaining practical experience with cGAN inference and computer vision.

References

Pix2Pix Paper, Official CycleGAN/Pix2Pix GitHub Repository, PyTorch Documentation.

Appendix 1

This appendix summarizes observations, advantages, limitations, and future improvements for the Pix2Pix model, including potential applications in image enhancement, medical imaging, satellite imagery, and architectural visualization.

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